Lagrange Multipliers

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1 Introduction

The method of Lagrange multipliers is a powerful technique for constrained optimization. Consider an objective function

$$f(x_1, x_2, \dots, x_n)$$

to be maximized or minimized subject to constraints of the form

$$g_i(x_1, x_2, \dots, x_n) = 0 \quad (i = 1, \dots, m).$$

Introduce auxiliary variables, the Lagrange multipliers λ_i, and define the auxiliary Lagrangian:

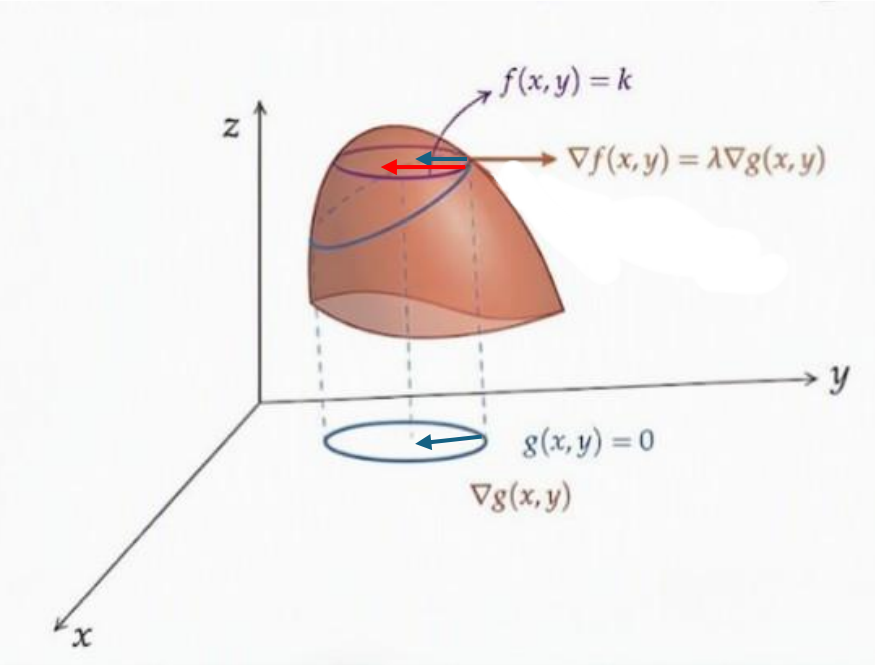
$$\mathcal{L}(x_1, \dots, x_n; \lambda_1, \dots, \lambda_m) = f(x_1, \dots, x_n) - \sum_{i=1}^{m} \lambda_i \, g_i(x_1, \dots, x_n).$$

At a regular constrained stationary point, where the constraint gradients are linearly independent, a necessary condition is

$$\nabla_x \mathcal{L} = 0, \qquad g_i(x) = 0.$$

Thus, the objective gradient ∇f is a linear combination of the constraint gradients ∇g_i. These equations identify candidates for extrema; they do not by themselves establish a maximum or minimum.

2 Geometric Interpretation of Lagrange Multipliers



Consider a function $f(x,y)$ subject to $g(x,y)=0$. The level curves of $f(x,y)$ join points where the function has the same value. The constraint $g(x,y)=0$ defines a curve in the plane. At a regular constrained extremum, a level curve of f is tangent to the constraint curve, provided the objective gradient is nonzero. Geometrically,

$$\nabla f(x,y) \parallel \nabla g(x,y),$$

the gradients are parallel, with λ giving their proportionality factor. A zero objective gradient is also compatible with the multiplier condition. The figure shows $g(x,y)=0$ in the xy plane and the surface $f(x,y)$. At a constrained maximum of $f(x,y)$ under $g(x,y)=0$, draw the level curve $f(x,y)=k$. Parameterizing it gives

$$r(t)=\big(x(t),y(t)\big), \quad f\big(x(t),y(t)\big)=k.$$

The chain rule gives

$$\frac{\partial f}{\partial x}\frac{dx}{dt}+\frac{\partial f}{\partial y}\frac{dy}{dt}=\left(\frac{\partial f}{\partial x},\frac{\partial f}{\partial y}\right)\cdot\left(\frac{dx}{dt},\frac{dy}{dt}\right)=\nabla f\cdot r'(t)=0.$$

Hence, the gradient of $f(x,y)$ is perpendicular to the tangent $r'(t)$: the gradient of f is normal to its level curve. At tangency, $\nabla f(x,y)$ and $\nabla g(x,y)$ are parallel. Equivalently, the derivative of the objective along the admissible tangent vanishes.

Example 1. Maximize $f(x,y)=x+y$ subject to $x^2+y^2=1$

- Constraint: $g(x,y)=x^2+y^2-1=0$.

- Lagrangian:

$$\mathcal{L} = x + y - \lambda(x^2 + y^2 - 1).$$

- Stationarity conditions:

$$\frac{\partial \mathcal{L}}{\partial x} = 1 - 2\lambda x = 0, \quad \frac{\partial \mathcal{L}}{\partial y} = 1 - 2\lambda y = 0.$$

- Solutions: $x = y = \pm 1/\sqrt{2}$, with matching signs for the two coordinates.
- Maximum: $f = \sqrt{2}$; minimum: $f = -\sqrt{2}$.

Example 2. Minimize $f(x, y) = x^2 + y^2$ subject to $x + y = 1$

- Constraint: $g(x, y) = x + y - 1 = 0$.
- Lagrangian:

$$\mathcal{L} = x^2 + y^2 - \lambda(x + y - 1).$$

- Stationarity conditions:

$$2x - \lambda = 0, \quad 2y - \lambda = 0 \quad \Rightarrow \quad x = y.$$

- The constraint gives $x = y = 1/2$.
- Minimum: $f = 1/2$. Restricting the objective to the constraint line gives a strictly convex quadratic, confirming the global minimum.

3 Lagrange Multipliers for Functionals

The same idea applies to functionals. Consider

$$J[y] = \int_a^b F\left(x, y(x), y'(x)\right) dx,$$

subject to an integral constraint:

$$G[y] = \int_a^b G\left(x, y(x), y'(x)\right) dx = 0.$$

Define the augmented functional

$$\mathcal{L}[y] = \int_a^b \left(F(x, y, y') - \lambda G(x, y, y')\right) dx.$$

Stationarity yields the modified Euler–Lagrange equation:

$$\frac{\partial}{\partial y}(F-\lambda G)-\frac{d}{dx}\left(\frac{\partial}{\partial y'}(F-\lambda G)\right)=0.$$

The multiplier λ is a constant enforcing this single integral constraint, assuming the constraint's first derivative is nonzero. This differs from a pointwise constraint, which generally requires a multiplier function of position.

In a function space, the first variations of $J[y]$ and $G[y]$ are linear functionals on admissible directions. At a regular constrained stationary point, these derivatives are proportional. A gradient representation of $\delta J[y]$ additionally requires a specified inner product and an appropriate representation theorem; a first variation is not itself a direction vector.

Example 1. Hohenberg–Kohn Energy

In the Kohn–Sham decomposition of density functional theory, the electronic energy is written as a functional of the electron density:

$$E[\rho]=T_s[\rho]+V_{ext}[\rho]+U[\rho]+E_{xc}[\rho].$$

Here, $T_s[\rho]$ is the noninteracting Kohn–Sham kinetic energy, $V_{ext}[\rho]$ is the external-potential energy, $U[\rho]$ is the classical Coulomb energy, and $E_{xc}[\rho]$ includes exchange, correlation, and the difference between interacting and noninteracting kinetic energies.

The exact Hohenberg-Kohn universal functional is the interacting kinetic plus electron-electron interaction energy. Its Kohn-Sham decomposition uses the noninteracting kinetic energy and absorbs the remaining kinetic and interaction effects into exchange and correlation:

$$F_{HK}[\rho]=T[\rho]+V_{ee}[\rho]=T_s[\rho]+U[\rho]+E_{xc}[\rho].$$

At fixed electron number, impose

$$\int \rho(\mathbf{r})\ d\mathbf{r}=N.$$

Construct the auxiliary functional

$$\mathcal{L}[\rho]=E[\rho]-\mu\left(\int \rho(\mathbf{r})\ d\mathbf{r}-N\right)$$

where μ is the multiplier associated with electron number, conventionally identified with the chemical potential. The Euler equation holds where the minimizing density is positive and differentiable; additional conditions apply at integer particle numbers and at the boundary of the admissible density set.

$$\frac{\delta\,\mathcal{L}[\rho]}{\delta\rho(\boldsymbol{r})}=\frac{\delta\,E}{\delta\rho(\boldsymbol{r})}-\mu=0.$$

$$\frac{\delta E}{\delta \rho(\boldsymbol{r})} = \mu.$$

Example 2. Kohn–Sham Equations

To obtain the Kohn-Sham orbitals, impose orthonormality:

$$\langle \phi_i | \phi_j \rangle = \delta_{ij}.$$

For occupied spin orbitals with unit occupation, in Hartree atomic units, the energy functional is

$$E[\{\phi_i\}] = \sum_i \langle \phi_i | -1/2\,\nabla^2 + V_{ext} | \phi_i \rangle + U[\rho] + E_{xc}[\rho].$$

$$\rho(r) = \sum\nolimits_{i=1}^{N} |\varphi_i(\boldsymbol{r})|^2.$$

The augmented functional is

$$\mathcal{L}[\phi_i, \epsilon_{ij}] = E[\phi_i] - \sum_{i,j} \epsilon_{ij} \left(\langle \phi_i | \phi_j \rangle - \delta_{ij} \right),$$

The matrix ϵ_{ij} enforces orbital orthonormality. The stationary equations initially couple the orbitals through this Hermitian multiplier matrix. A unitary rotation within the occupied space diagonalizes it, yielding canonical Kohn-Sham orbitals.

$$H_{KS} = -\frac{1}{2}\nabla^2 + v_{ext}(\boldsymbol{r}) + v_H(\boldsymbol{r}) + v_{xc}(\boldsymbol{r}),$$

$$v_H(r) = \frac{\delta U}{\delta \rho(\boldsymbol{r})}, \quad v_{xc}(\boldsymbol{r}) = \frac{\delta E_{xc}}{\delta \rho(\boldsymbol{r})}$$

$$H_{KS}\varphi_i = \sum\nolimits_j \varepsilon_{ij}\varphi_j.$$

The resulting canonical equations are

$$\widehat{H}_{KS}\phi_i = \epsilon_i \phi_i.$$